

$$\begin{aligned} 0.3667 \varphi_2 - 0.2 \varphi_3 - 0.1 \varphi_4 &= 2.6667 \\ -0.2 \varphi_2 + 0.4778 \varphi_3 - 0.1667 \varphi_4 &= -18.2222 \\ -0.1 \varphi_2 - 0.1667 \varphi_3 + 0.3167 \varphi_4 &= 16.6667 \end{aligned}$$

Где

$$\begin{aligned} G_3 &= \frac{1}{R_3} = \frac{1}{15} = 0.066667 \text{ S} \\ G_2 &= \frac{1}{R_2} = \frac{1}{9} = 0.111111 \text{ S} \\ G_5 &= \frac{1}{R_5} = \frac{1}{20} = 0.05 \text{ S} \\ G_1 &= \frac{1}{R_1} = \frac{1}{6} = 0.166667 \text{ S} \\ G_6 &= \frac{1}{R_6} = \frac{1}{10} = 0.1 \text{ S} \end{aligned}$$

Решим систему с помощью матриц

$$\varphi = (G_B^{-1})^T (J_B - G_B E_B)$$

После расчета получим

$$\varphi_2 = 10.551 \text{ V}$$

$$\varphi_3 = -17.394 \text{ V}$$

$$\varphi_4 = 46.809 \text{ V}$$

Определяем токи ветвей

$$I_3 = \frac{v_3 - v_2}{R_3} = \frac{0 - (10.551) - 100}{15} = -5.963 \text{ A}$$

$$I_2 = \frac{v_3 - v_1}{R_2} = \frac{(-17.394) - (0) - 50}{9} = -3.623 \text{ A}$$

$$I_5 = \frac{v_2 - v_1}{R_5} = \frac{46.809 - (0)}{20} = 2.34 \text{ A}$$

$$I_1 = \frac{v_3 - v_2}{R_1} = \frac{(-17.394) - (10.551) - 100}{6} = -5.966 \text{ A}$$

$$I_6 = \frac{v_2 - v_4}{R_6} = \frac{46.809 - (10.551)}{10} = 3.626 \text{ A}$$

$$I_0 = (\varphi_3 - \varphi_2)G_0 = (-17.394 - (10.551)) \cdot 0.2 = -5.589 \text{ A}$$

$$J_0 = J_0 = 4 \text{ A}$$